Revised simplex algorithm

Outline

Pivoting operations with two systems of linear equations

Pseudo-code of the revised simplex algorithm

The revised simplex algorithm

- The simplex algorithm moves from a basic feasible solution to another one with **pivoting operations**
- A pivoting operation needs: the **reduced costs**, the basic feasible solution **value**, the column of $\bar{\boldsymbol{A}}$ of the **pivot column** x_p and the values of the basic variables
- The **revised simplex algorithm** computes only this information, by solving **two systems of linear equations** with the basis matrix $\boldsymbol{A}_{\mathcal{B}}$
- We follow it on an LP problem with 2 variables and 3 constraints

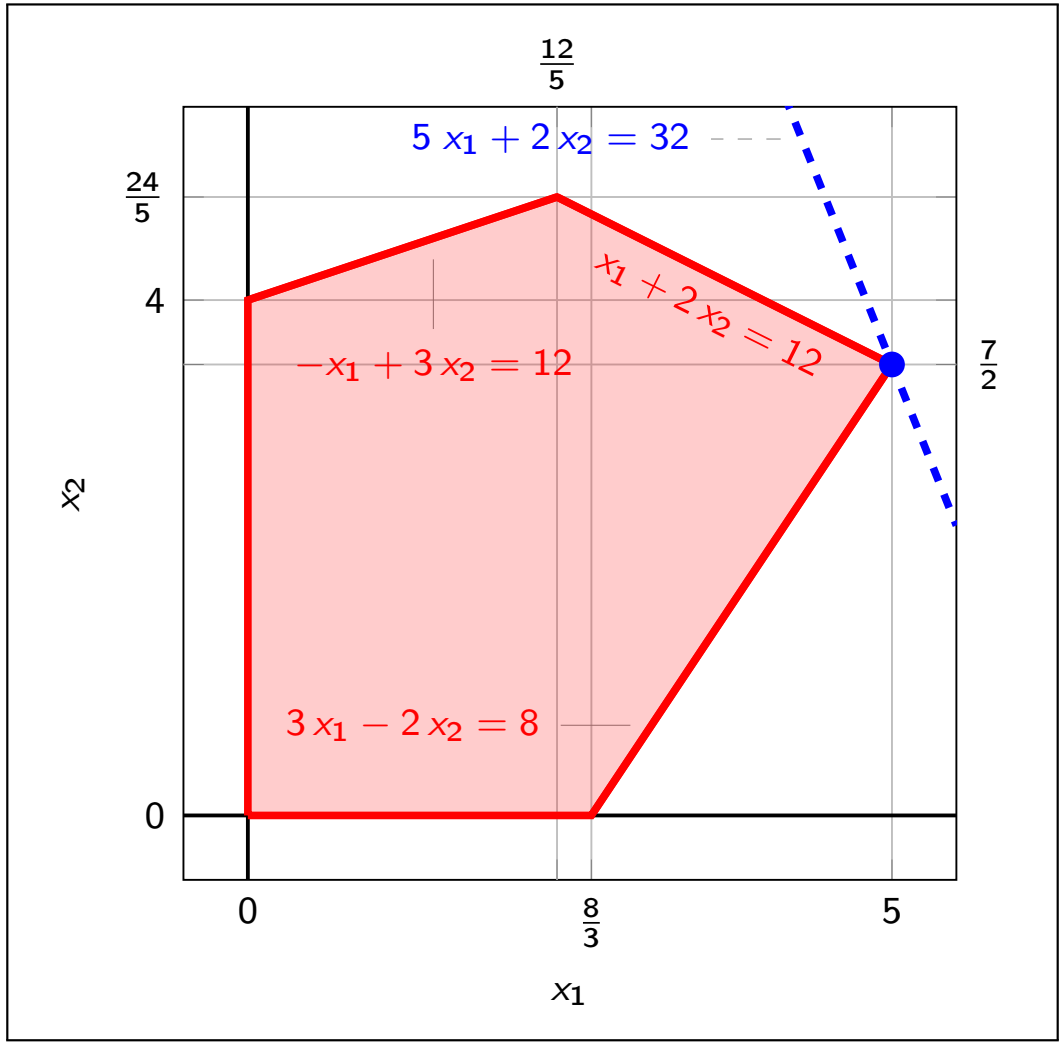
Example: canonical form and standard form (1/5)

We consider the following LP problem in **canonical maximization form** with $n = 2$ variables and $m = 3$ constraints:

$$
\begin{array}{rrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

- In the next slide: its feasible region in the plane (x_1, x_2)

Example: canonical form and standard form (2/5)



- The blue point $(5, \frac{7}{2})$ lies on the dashed level line $5\,x_1 + 2\,x_2 = 32$

Example: canonical form and standard form (3/5)

Its equivalent **standard form** with $n + m = 5$ variables and $m = 3$ constraints is:

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & & & & & & & & \\
\text{subject to} & -x_1 & + & 3 & x_2 & + & x_3 & & & & & = & 12 \\
& x_1 & + & 2 & x_2 & & & + & x_4 & & & = & 12 \\
& 3 & x_1 & - & 2 & x_2 & & & & + & x_5 & = & 8 \\
& x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

Example: canonical form and standard form (4/5)

For this LP problem in standard form, we have:

$$\boldsymbol{c} = \begin{pmatrix} 5 \\ 2 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} -1 & 3 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 3 & -2 & 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

Example: canonical form and standard form (5/5)

Since $\begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \geq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, with $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$ we have the following initial basic feasible solution:

$$\begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \text{and} \qquad \bar{\boldsymbol{b}} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

Pivoting operations with two systems of linear equations

Introduction: Pivoting operations with two systems of linear equations

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$, all the information necessary for the pivoting operations can be determined by solving **two systems of linear equations**
- The **first system** gives the reduced costs and the basic feasible solution value: the optimality check and the **pivot column**
- The **second system** gives the **pivot row** and the new basic feasible solution
- Both systems have the same matrix $\boldsymbol{A}_{\mathcal{B}}$, and the inverse $\boldsymbol{A}_{\mathcal{B}}^{-1}$ is never computed

Recall: the quantities of a basis

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$, with basic indices $\mathcal{B}$ and non-basic indices $\mathcal{N}$, we use (as earlier in the course):

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \qquad \bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}$$

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \qquad \bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}$$

- All of them contain the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$: we now see how to obtain what we need without computing it

First system of linear equations (1/3)

- The reduced costs and the basic feasible solution value can be determined by computing directly $\boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$ since:

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}\ \boldsymbol{A}_{\mathcal{N}}$$

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}\ \boldsymbol{b}$$

- The same row vector $\boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$ appears in both formulas

First system of linear equations (2/3)

- By setting $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$, the row vector $\boldsymbol{\pi} \in \mathbb{R}^{1 \times m}$ of m columns can be found by solving the following system of linear equations:

$$\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}} \quad \Longrightarrow \quad \boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$$

$$\text{where} \quad \boldsymbol{\pi} = [\pi_j] \ \text{ with } \ j \in \mathcal{B}$$

First system of linear equations (3/3)

- In the examples the components of π are labelled with the indices $j \in \mathcal{B}$ of the basic variables
- But each component of π corresponds to a constraint, i.e., to a row of $\boldsymbol{A}_{\mathcal{B}}$
- The i-th component of π is associated with the i-th constraint

From the pivot column to the pivot row

The first system tells us whether the basis is optimal and which non-basic variable can enter the basis.

Once the pivot column is known, a second system gives the pivot row and the new basic feasible solution.

Second system of linear equations (1/2)

- Given a pivot column x_p with $p \in \mathcal{N}$, the pivot row and the update of the current basic feasible solution can be done by directly computing $\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{A}_p$ since:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{A}_p\,x_p$$

Second system of linear equations (2/2)

- By setting $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{A}_p$, the column vector $\boldsymbol{d} \in \mathbb{R}^{m\times 1}$ of m rows (associated with the basic variables) can be found by solving the following system of linear equations:

$$\boldsymbol{A}_{\mathcal{B}}\ \boldsymbol{d} = \boldsymbol{A}_p \quad \Longrightarrow \quad \boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{A}_p$$

$$\text{where} \quad \boldsymbol{d} = [d_i] \ \text{ with } \ i \in \mathcal{B}$$

The two systems on the example

Let us solve the two systems on the LP problem of the example,
with the basis $\mathcal{B} = \{1, 3, 4\}$
and the pivot column x_2.

Example: systems of linear equations (1/3)

For the LP problem of the example above with $\mathcal{B} = \{1, \ 3, \ 4\}$, $\mathcal{N} = \{2, \ 5\}$ and x_2 as the pivot column, we have:

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix} \qquad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} \qquad \boldsymbol{A}_2 = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

- The columns of $\boldsymbol{A}_{\mathcal{B}}$ are the columns $\boldsymbol{A}_1$, $\boldsymbol{A}_3$, $\boldsymbol{A}_4$ of $\boldsymbol{A}$

Example: systems of linear equations (2/3)

We solve $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ to compute $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$,
first system of linear equations:

$$\begin{cases} -\pi_1 + \pi_3 + 3\,\pi_4 = 5 \\ \pi_1 = 0 \\ \pi_3 = 0 \end{cases} \quad \Rightarrow \quad \boldsymbol{\pi} = \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix}$$

Example: systems of linear equations (3/3)

We solve $\boldsymbol{A}_{\mathcal{B}}\,\boldsymbol{d} = \boldsymbol{A}_2$ to compute $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{A}_2$,
second system of linear equations:

$$\begin{cases} -d_1 + d_3 = 3 \\ d_1 + d_4 = 2 \\ 3\,d_1 = -2 \end{cases} \Rightarrow \quad \boldsymbol{d} = \begin{pmatrix} -\frac{2}{3} \\ \frac{7}{3} \\ \frac{8}{3} \end{pmatrix}$$

Two systems instead of an inverse matrix

- The revised simplex algorithm never computes the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$
- At every iteration it solves two systems of linear equations: one for the row vector $\boldsymbol{\pi}$ and one for the column vector $\boldsymbol{d}$
- Solving a system of m linear equations is computationally cheaper than inverting a matrix of size m
- The matrix $\boldsymbol{A}_{\mathcal{B}}$ of the two systems is the same

Summary

First system of linear equations:

- $\pi\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ gives $\pi = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$
- Then $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \pi\ \boldsymbol{A}_{\mathcal{N}}$ and $\bar{z} = \pi\ \boldsymbol{b}$

Second system of linear equations:

- $\boldsymbol{A}_{\mathcal{B}}\ \boldsymbol{d} = \boldsymbol{A}_p$ gives $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{A}_p$, the column of $\bar{\boldsymbol{A}}$ of the pivot column x_p
- Then $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \boldsymbol{d}\ x_p$

No inverse matrix: two systems with the same matrix $\boldsymbol{A}_{\mathcal{B}}$

Pseudo-code of the revised simplex algorithm

Introduction: Pseudo-code of the revised simplex algorithm

- We put the two systems of linear equations together in the **pseudo-code** of the revised simplex algorithm
- Three steps: **initialization**, **optimality check** and **pivoting operations**, repeated until an optimal solution is found or the LP problem is unbounded
- Then we run the algorithm on the LP problem of the example: **three iterations** up to the optimal solution

Revised simplex algorithm (1/4)

Step 0: **Initialization**

Find a basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ of the LP problem in standard maximization form. Go to Step 1.

Step 1: **Optimality check**

a. Solve $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ to compute $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$

b. Compute the reduced costs and the basic feasible solution value:

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \underbrace{\boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=\boldsymbol{\pi}}\ \boldsymbol{A}_{\mathcal{N}} \qquad \bar{z} = \underbrace{\boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=\boldsymbol{\pi}}\ \boldsymbol{b}$$

If $\bar{\boldsymbol{c}}' \leq 0$ then the basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ is an optimal solution and $\bar{z}$ is the optimal value **else** go to Step 2.

Revised simplex algorithm (2/4)

Step 2: **Pivoting operations**

The **pivot column** is a non-basic variable x_p with $p \in \mathcal{N}$ such that:

$$c_p > \underbrace{\boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1}}_{=\boldsymbol{\pi}} \, \boldsymbol{A}_p$$

a. Solve $\boldsymbol{A}_{\mathcal{B}} \, \boldsymbol{d} = \boldsymbol{A}_p$ to compute $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p$
b. If $\boldsymbol{d} \leq 0$, then the LP problem is **unbounded**. STOP.
c. Find the largest value t (called t_m) such that:

$$\bar{\boldsymbol{b}} - t \, \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p}_{=\boldsymbol{d}} \geq 0$$

$$\Longrightarrow \quad t_m \leftarrow \min \left\{ \frac{\bar{b}_i}{d_i} : i \in \mathcal{B}, \; d_i > 0 \right\}$$

Revised simplex algorithm (3/4)

The **pivot row** is a basic variable x_r with $r \in \mathcal{B}$ such that:

$$\bar{b}_r - t_m \underbrace{\bar{a}_{rp}}_{=d_r} = 0$$

d. Update the basis with the new partition of the indices of the variables:

$$\mathcal{B} \leftarrow \left(\mathcal{B} \setminus \{r\}\right) \cup \{p\} \quad \text{and} \quad \mathcal{N} \leftarrow \left(\mathcal{N} \setminus \{p\}\right) \cup \{r\}$$

Revised simplex algorithm (4/4)

e. Update the basic feasible solution (new vector $\bar{\boldsymbol{b}}$):

$$x_i = \bar{b}_i - t_m \underbrace{\bar{a}_{ip}}_{=d_i}, \ \forall i \in \mathcal{B} \setminus \{p\}, \quad x_p = t_m$$

$$\text{and} \quad x_i = 0, \ \forall i \in \mathcal{N}$$

Go to Step 1.

Reading the pseudo-code

- **Step 1** solves the first system: π gives at once the reduced costs and the value $\bar{z}$
- **Step 2** solves the second system: $\boldsymbol{d}$ gives the unboundedness test, the largest step t_m and the pivot row
- Each iteration changes only one index of $\mathcal{B}$: the matrix $\boldsymbol{A}_{\mathcal{B}}$ of the next iteration has only one column different from the current one

The algorithm on the example

We now run the revised simplex algorithm
on the LP problem of the example,
starting from $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$.

Each iteration: optimality check (first system),
then pivoting operation (second system).

Example: first iteration (1/6)

Initialization

In the LP problem of the example above,

$$\begin{pmatrix} x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

is a basic feasible solution with

$$\bar{\boldsymbol{b}} = \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix}$$

Example: first iteration (2/6)

Optimality check

With $\mathcal{B} = \{3, 4, 5\}$ and $\mathcal{N} = \{1, 2\}$, we solve $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ to compute $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$.

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

$$\text{then} \quad \begin{cases} \pi_3 = 0 \\ \pi_4 = 0 \\ \pi_5 = 0 \end{cases} \quad \Rightarrow \quad \boldsymbol{\pi} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$$

Example: first iteration (3/6)

We compute the basic feasible solution value:

$$\bar{z} = \boldsymbol{\pi}\ \boldsymbol{b} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = 0$$

We compute the reduced costs $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{N}}$:

$$\bar{\boldsymbol{c}}' = \begin{pmatrix} 5 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 3 & -2 \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 5 & 2 \end{pmatrix}}_{c_1 \quad c_2} - \underbrace{\begin{pmatrix} 0 & 0 \end{pmatrix}}_{\boldsymbol{\pi}\ \boldsymbol{A}_1 \quad \boldsymbol{\pi}\ \boldsymbol{A}_2} = \underbrace{\begin{pmatrix} 5 & 2 \end{pmatrix}}_{\bar{c}_1 \quad \bar{c}_2}$$

Example: first iteration (4/6)

Pivoting operation

Since $c_1 > \pi\, \boldsymbol{A}_1$, x_1 is the pivot column. We solve $\boldsymbol{A}_{\mathcal{B}}\, \boldsymbol{d} = \boldsymbol{A}_1$ to compute $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_1$.

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{A}_1 = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

$$\text{then} \quad \begin{cases} d_3 = -1 \\ d_4 = 1 \\ d_5 = 3 \end{cases} \quad \Rightarrow \quad \boldsymbol{d} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

Example: first iteration (5/6)

We have to find the largest t such that $\bar{\boldsymbol{b}} - t\,\boldsymbol{d} \geq 0$:

$$\begin{cases} \underbrace{12 - t \cdot (-1) \geq 0}_{x_3 \geq 0} \\ \underbrace{12 - t \cdot 1 \geq 0}_{x_4 \geq 0} \\ \underbrace{8 - t \cdot 3 \geq 0}_{x_5 \geq 0} \end{cases} \qquad \begin{cases} t \geq -12 \\ t \leq 12 \\ t \leq \frac{8}{3} \end{cases} \qquad \text{then} \quad t_m = \frac{8}{3}$$

- The value of the pivot column becomes t_m, i.e., $x_1 = \frac{8}{3}$
- The value of x_5 becomes zero and x_5 is the pivot row

Example: first iteration (6/6)

The other basic variables take values $\bar{\boldsymbol{b}} - t_m \, \boldsymbol{d}$, then we have:

$$x_3 = 12 - \frac{8}{3} \cdot (-1) = \frac{44}{3}, \quad x_4 = 12 - \frac{8}{3} \cdot 1 = \frac{28}{3}$$

and accordingly

$$\begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \frac{8}{3} \\ \frac{44}{3} \\ \frac{28}{3} \end{pmatrix}$$

The new partition of the indices of the variables is $\mathcal{B} = \{1, 3, 4\}$ and $\mathcal{N} = \{2, 5\}$.

After the first iteration

- x_1 has entered the basis and x_5 has left it: $\mathcal{B} = \{1, 3, 4\}$, $\mathcal{N} = \{2, 5\}$
- The matrix $\boldsymbol{A}_{\mathcal{B}}$ now has the columns $\boldsymbol{A}_1$, $\boldsymbol{A}_3$, $\boldsymbol{A}_4$: it is the basis of the systems we solved in the example of the previous section
- We repeat the optimality check with the new basis

Example: second iteration (1/5)

Optimality check

With $\mathcal{B} = \{1, 3, 4\}$ and $\mathcal{N} = \{2, 5\}$, we solve $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ to compute $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$.

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix} \quad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 5 & 0 & 0 \end{pmatrix}$$

$$\text{then} \quad \begin{cases} -\pi_1 + \pi_3 + 3\,\pi_4 = 5 \\ \pi_1 = 0 \\ \pi_3 = 0 \end{cases} \quad \Rightarrow \quad \boldsymbol{\pi} = \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix}$$

Example: second iteration (2/5)

We compute the basic feasible solution value:

$$\bar{z} = \boldsymbol{\pi}\ \boldsymbol{b} = \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = \frac{40}{3}$$

We compute the reduced costs $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{N}}$:

$$\bar{\boldsymbol{c}}' = \begin{pmatrix} 2 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & \frac{5}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 2 & 0 \\ -2 & 1 \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 2 & 0 \end{pmatrix}}_{c_2 \quad c_5} - \underbrace{\begin{pmatrix} -\frac{10}{3} & \frac{5}{3} \end{pmatrix}}_{\boldsymbol{\pi}\ \boldsymbol{A}_2 \quad \boldsymbol{\pi}\ \boldsymbol{A}_5} = \underbrace{\begin{pmatrix} \frac{16}{3} & -\frac{5}{3} \end{pmatrix}}_{\bar{c}_2 \quad \bar{c}_5}$$

Example: second iteration (3/5)

Pivoting operation

Since $c_2 > \boldsymbol{\pi}\, \boldsymbol{A}_2$, x_2 is the pivot column. We solve $\boldsymbol{A}_{\mathcal{B}}\, \boldsymbol{d} = \boldsymbol{A}_2$ to compute $\boldsymbol{d} = \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_2$.

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix} \quad \boldsymbol{A}_2 = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

$$\text{then} \quad \begin{cases} -d_1 + d_3 = 3 \\ d_1 + d_4 = 2 \\ 3\, d_1 = -2 \end{cases} \quad \Rightarrow \quad \boldsymbol{d} = \begin{pmatrix} -\frac{2}{3} \\ \frac{7}{3} \\ \frac{8}{3} \end{pmatrix}$$

Example: second iteration (4/5)

We have to find the largest t such that $\bar{\boldsymbol{b}} - t\,\boldsymbol{d} \geq 0$:

$$\begin{cases} \underbrace{\frac{8}{3} - t \cdot \left(-\frac{2}{3}\right) \geq 0}_{x_1 \geq 0} \\ \underbrace{\frac{44}{3} - t \cdot \frac{7}{3} \geq 0}_{x_3 \geq 0} \\ \underbrace{\frac{28}{3} - t \cdot \frac{8}{3} \geq 0}_{x_4 \geq 0} \end{cases} \qquad \begin{cases} t \geq -4 \\ t \leq \frac{44}{7} \\ t \leq \frac{7}{2} \end{cases} \qquad \text{then} \quad t_m = \frac{7}{2}$$

- The value of the pivot column becomes t_m, i.e., $x_2 = \frac{7}{2}$
- The value of x_4 becomes zero and x_4 is the pivot row

Example: second iteration (5/5)

The other basic variables take values $\bar{\boldsymbol{b}} - t_m \, \boldsymbol{d}$, then we have:

$$x_1 = \frac{8}{3} - \frac{7}{2} \cdot \left(-\frac{2}{3} \right) = 5, \quad x_3 = \frac{44}{3} - \frac{7}{2} \cdot \frac{7}{3} = \frac{13}{2}$$

and accordingly

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{7}{2} \\ \frac{13}{2} \end{pmatrix}$$

The new partition of the indices of the variables is $\mathcal{B} = \{1, 2, 3\}$ and $\mathcal{N} = \{4, 5\}$.

After the second iteration

- x_2 has entered the basis and x_4 has left it: $\mathcal{B} = \{1, 2, 3\}$, $\mathcal{N} = \{4, 5\}$
- The basic feasible solution of the second iteration has value $\bar{z} = \frac{40}{3}$: let us see whether the new one is optimal
- We repeat the optimality check with the new basis

Example: third iteration (1/3)

Optimality check

With $\mathcal{B} = \{1, 2, 3\}$ and $\mathcal{N} = \{4, 5\}$, we solve $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$ to compute $\boldsymbol{\pi} = \boldsymbol{c}'_{\mathcal{B}}\ \boldsymbol{A}_{\mathcal{B}}^{-1}$.

$$\boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} -1 & 3 & 1 \\ 1 & 2 & 0 \\ 3 & -2 & 0 \end{pmatrix} \quad \boldsymbol{c}'_{\mathcal{B}} = \begin{pmatrix} 5 & 2 & 0 \end{pmatrix}$$

$$\text{then} \quad \begin{cases} -\pi_1 + \pi_2 + 3\,\pi_3 = 5 \\ 3\,\pi_1 + 2\,\pi_2 - 2\,\pi_3 = 2 \\ \pi_1 = 0 \end{cases} \quad \Rightarrow \quad \boldsymbol{\pi} = \begin{pmatrix} 0 & 2 & 1 \end{pmatrix}$$

Example: third iteration (2/3)

We compute the basic feasible solution value:

$$\bar{z} = \boldsymbol{\pi}\ \boldsymbol{b} = \begin{pmatrix} 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 12 \\ 12 \\ 8 \end{pmatrix} = 32$$

We compute the reduced costs $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{N}}$:

$$\bar{\boldsymbol{c}}' = \begin{pmatrix} 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} 0 & 0 \end{pmatrix}}_{c_4 \quad c_5} - \underbrace{\begin{pmatrix} 2 & 1 \end{pmatrix}}_{\boldsymbol{\pi}\ \boldsymbol{A}_4 \quad \boldsymbol{\pi}\ \boldsymbol{A}_5} = \underbrace{\begin{pmatrix} -2 & -1 \end{pmatrix}}_{\bar{c}_4 \quad \bar{c}_5}$$

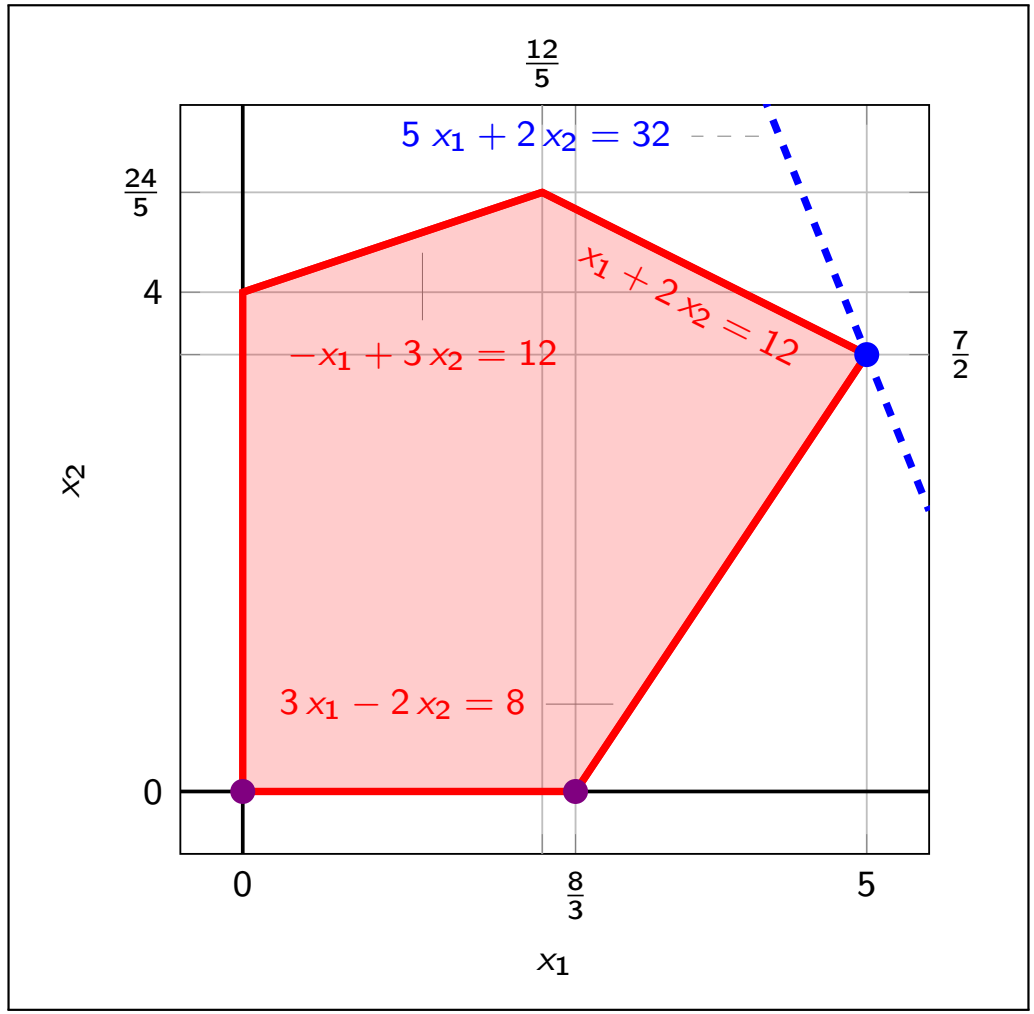
Example: third iteration (3/3)

All reduced costs are non-positive, then the basic feasible solution

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{7}{2} \\ \frac{13}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

is an optimal solution of the LP problem and $\bar{z} = 32$ is the optimal value.

The three basic feasible solutions in the plane



- In (x_1, x_2): from $(0, 0)$ to $(\frac{8}{3}, 0)$ (violet) and then to the optimal vertex $(5, \frac{7}{2})$ (blue)

Summary

Pseudo-code:

- Step 0: a basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$, $\boldsymbol{x}_{\mathcal{N}} = 0$
- Step 1: first system for $\boldsymbol{\pi}$, then $\bar{\boldsymbol{c}}'$ and $\bar{z}$; stop if $\bar{\boldsymbol{c}}' \leq 0$
- Step 2: second system for $\boldsymbol{d}$; unbounded if $\boldsymbol{d} \leq 0$, otherwise $t_m = \min\{\bar{b}_i/d_i : i \in \mathcal{B},\ d_i > 0\}$ and new partition $\mathcal{B}$, $\mathcal{N}$

On the example:

- Bases $\{3,4,5\} \to \{1,3,4\} \to \{1,2,3\}$, values $0 \to \frac{40}{3} \to 32$
- Optimal solution $x_1 = 5$, $x_2 = \frac{7}{2}$, $x_3 = \frac{13}{2}$, optimal value 32

Summary of the chapter (1/2)

The revised simplex algorithm:

- It performs the same pivoting operations as the simplex algorithm
- It never computes the inverse matrix $\boldsymbol{A}_{\mathcal{B}}^{-1}$

Two systems of linear equations with the same matrix $\boldsymbol{A}_{\mathcal{B}}$:

- $\boldsymbol{\pi}\ \boldsymbol{A}_{\mathcal{B}} = \boldsymbol{c}'_{\mathcal{B}}$: the row vector $\boldsymbol{\pi}$, one component for each constraint
- $\boldsymbol{A}_{\mathcal{B}}\ \boldsymbol{d} = \boldsymbol{A}_p$: the column vector $\boldsymbol{d}$, one component for each basic variable

Summary of the chapter (2/2)

From π:

- Reduced costs $\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{\pi}\, \boldsymbol{A}_{\mathcal{N}}$ and value $\bar{z} = \boldsymbol{\pi}\, \boldsymbol{b}$
- Optimality check and pivot column x_p with $c_p > \boldsymbol{\pi}\, \boldsymbol{A}_p$

From $\boldsymbol{d}$:

- Unboundedness test $\boldsymbol{d} \leq 0$
- Largest step t_m, pivot row x_r and new basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - t_m\, \boldsymbol{d}$, $x_p = t_m$